Beyond Black–Scholes

Lecture 12 Notes — Local Volatility, Stochastic Volatility, and Stochastic Rates

Goals and logical structure

A useful model must explain the prices and risks that matter:

$$\text{implied-volatility surface} \longrightarrow \text{local volatility} \longrightarrow \text{random volatility} \longrightarrow \text{random discounting}.$$

The trade-off is fit, realism, hedging, and tractability.

1 Why Black–Scholes is not enough

Under Black–Scholes,

$$\frac{\mathrm{d}S_t}{S_t} = r\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}}, \qquad \sigma \text{ constant}.$$

If this were exact, implied volatility would be constant across strikes and maturities. Markets instead quote a surface $(K,T) \mapsto \sigma_{\mathrm{imp}}(K,T)$, with smiles, skews, and term structure.

Example — equity skew

For a fixed maturity, equity implied volatility typically satisfies

$$\sigma_{\mathrm{imp}}(K_{\mathrm{low}}) > \sigma_{\mathrm{imp}}(K_{\mathrm{ATM}}) > \sigma_{\mathrm{imp}}(K_{\mathrm{high}}).$$

Low-strike puts insure severe declines; their demand and price are high. Implied volatility is the Black–Scholes parameter matching a quote, not a literal historical volatility of that option.

Three extensions, three questions

Observed issue	Extension	Main purpose
Static vanilla smile	local volatility	fit prices today
Volatility dynamics	stochastic volatility	model future smile and vol risk
Yield-curve risk	stochastic rates	model bonds and discounting

2 Local volatility fits today's surface

2.1 One-factor state-dependent variance

Definition 2.1 (Local-volatility model)**.** Under $\mathbb{Q}$,

$$\frac{\mathrm{d}S_t}{S_t} = r\,\mathrm{d}t + \sigma_{\mathrm{loc}}(t,S_t)\,\mathrm{d}W_t.$$

The instantaneous variance depends deterministically on time and current spot. The model remains one-factor Markovian.

For European payoff $g(S_T)$, the price solves

$$u_t + rsu_s + \frac{1}{2}\sigma_{\mathrm{loc}}(t,s)^2 s^2 u_{ss} - ru = 0, \qquad u(T,s) = g(s).$$

Only the constant variance of Black–Scholes has been replaced.

2.2 Dupire calibration

With constant rates and no dividends, an arbitrage-free call-price surface $C(T,K)$ gives formally

$$\boxed{\sigma_{\mathrm{loc}}(T,K)^2 = \frac{C_T(T,K) + rKC_K(T,K)}{\frac{1}{2}K^2 C_{KK}(T,K)}.}$$

Strike convexity is crucial:

$$C_{KK}(T,K) = e^{-rT} p^{\mathbb{Q}}_{S_T}(K) \geq 0.$$

Differentiate a clean surface, not raw quotes

Dupire uses a second strike derivative. Noise in a sparse quote table is amplified and can give negative or unstable local variance. First build a smooth, arbitrage-free price or implied-volatility surface.

Exercise — local or extra state?

Does local volatility introduce an additional random factor besides S_t?

Short solution

No. $\sigma_{\mathrm{loc}}(t,S_t)$ is a deterministic function of the current time and stock. The state remains (t,S_t).

Strength and limitation

Local volatility can reproduce a clean grid of vanilla prices today. It also prescribes future smile motion through a deterministic spot-time rule; that future dynamics need not be realistic for exotic hedging.

3 Stochastic volatility makes variance a risk factor

3.1 A two-factor model

A generic specification is

$$\frac{\mathrm{d}S_t}{S_t} = r\,\mathrm{d}t + \sqrt{V_t}\,\mathrm{d}W_t^{(1)}, \qquad \mathrm{d}V_t = b(t,V_t)\,\mathrm{d}t + a(t,V_t)\,\mathrm{d}W_t^{(2)},$$

$$\mathrm{d}\langle W^{(1)}, W^{(2)}\rangle_t = \rho\,\mathrm{d}t.$$

The Markov state is now (S_t, V_t). Two paths with the same spot can have different option values if their current variance levels differ.

Example — Heston

$$\frac{\mathrm{d}S_t}{S_t} = r\,\mathrm{d}t + \sqrt{V_t}\,\mathrm{d}W_t^{(1)}, \qquad \mathrm{d}V_t = \kappa(\theta - V_t)\,\mathrm{d}t + \eta\sqrt{V_t}\,\mathrm{d}W_t^{(2)}.$$

θ is the long-run variance, κ mean-reversion speed, η volatility of volatility, and ρ spot-variance correlation.

The conditional mean illustrates mean reversion:

$$\mathbb{E}[V_T \mid V_t = v] = \theta + (v - \theta)e^{-\kappa(T-t)}.$$

In equities, $\rho < 0$ captures the leverage effect: a negative stock shock tends to accompany higher variance, making downside puts relatively expensive and generating skew.

Richer dynamics imply hedging risk

There are two sources of randomness but often only stock and cash are traded. The model is generally incomplete: stock delta alone cannot perfectly hedge variance risk. Liquid vanilla options are often used as additional risk-management instruments.

Exercise — choose a volatility model

Which model is more natural for a payoff sensitive to future realized volatility: local or stochastic volatility?

Short solution

Stochastic volatility, because it retains a random variance state whose future dynamics can affect the payoff and hedge.

4 Stochastic rates make discounting random

4.1 Bonds and the yield curve

With short rate r_t,

$$D(t,T) = \exp\left(-\int_t^T r_s \, \mathrm{d}s\right), \qquad P(t,T) = \mathbb{E}^{\mathbb{Q}}[D(t,T) \mid \mathcal{F}_t].$$

The curve can be parameterized equivalently by

$$P(t,T) = e^{-R(t,T)(T-t)}, \qquad R(t,T) = -\frac{\log P(t,T)}{T-t}, \qquad f(t,T) = -\partial_T \log P(t,T).$$

Example — a curve calculation

If $P(0,1) = .96$ and $P(0,2) = .90$, then

$$R(0,2) = -\tfrac{1}{2}\log(.90) = 5.27\%, \qquad L(0;1,2) = \frac{.96}{.90} - 1 = 6.67\%.$$

The second-year forward rate exceeds the two-year average zero rate.

4.2 Short-rate and forward-rate models

A short-rate model specifies r_t and derives bond prices. Examples:

$$\mathrm{d}r_t = a(b - r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t \quad \text{(Vasicek)},$$

$$\mathrm{d}r_t = a(b - r_t)\,\mathrm{d}t + \sigma\sqrt{r_t}\,\mathrm{d}W_t \quad \text{(CIR)}.$$

HJM starts directly from the forward curve,

$$\mathrm{d}f(t,T) = \alpha(t,T)\,\mathrm{d}t + \sigma(t,T)\,\mathrm{d}W_t.$$

Under the risk-neutral measure, no-arbitrage determines the drift α once forward-rate volatility is chosen.

Exercise — when are rates material?

Is deterministic discounting a more defensible approximation for a one-month vanilla equity option or for a five-year callable bond?

Short solution

The short-dated equity option is the more defensible case. A callable bond is directly exposed to the future yield curve and an exercise decision, so stochastic rates are material.

5 Calibration and model choice

Calibration is a market-pricing step

Local volatility fits an arbitrage-free vanilla surface. Stochastic volatility fits it through parameters such as $\kappa, \theta, \eta, \rho, V_0$. Rate models fit the initial curve together with liquid caps, swaptions, or bond-option prices. Calibration aligns a model with tradable prices before it is used for valuation or risk.

Use the least complex model that captures material risk

Local volatility is a natural first benchmark for accurate current vanilla fit. A long-dated equity-linked payoff with rate exposure may need stochastic volatility and stochastic rates. More factors mean more calibration burden and more hedging uncertainty.

Exercise — final model choice

A desk must hedge a barrier option and reproduce the current vanilla smile surface accurately. What is the natural first benchmark, and what caveat remains?

Short solution

Local volatility is the natural first benchmark because it fits today's vanilla surface. Its future smile dynamics can be unrealistic, so hedge performance remains model dependent.

Final synthesis

$$\boxed{\text{Market prices} \longrightarrow \text{model assumptions} \longrightarrow \text{calibration} \longrightarrow \text{pricing and hedging.}}$$

A richer model earns its place only by explaining a material source of risk.